

The Conversion Engine — Decision Memo

To: Tenacious CEO & CFO From: Chalie Lijam · TRP1 Week 10 Date: 2026-04-26 Repo: The_Conversion_Engine / main

1. EXECUTIVE SUMMARY

The Conversion Engine is a fully automated B2B outreach pipeline that enriches prospects from three public data sources (Crunchbase, layoffs.fyi, careers pages), classifies them into four ICP segments with a confidence-gated LLM, and syncs every event to HubSpot in real time. On τ^2 -Bench retail, the model passed 21/21 tasks it actually executed (pass@1 = 1.00 on executed tasks; 0.70 overall with 9 API-quota infrastructure errors); on Tenacious-specific probes it cut bench over-commitment by 42pp (p<0.001) and tone non-compliance by 30pp (p=0.003), at \$0.015 per qualified lead — 333× under the \$5 hard cap. We recommend a 30-day Segment 1 pilot at 50 leads/week.

2. COST PER QUALIFIED LEAD

Qualified lead defined: A prospect classified into a non-abstained ICP segment (confidence ≥ 0.40) who receives a signal-grounded email passing tone-guard validation. Abstained prospects (~18% of pipeline) are excluded.

Component	Source	Amount
LLM composition (150 baseline sims)	eval/invoice_summary.json · eval/trace_log.jsonl	\$2.99 total · \$0.0199 avg/sim
Tone guard (Claude Haiku 4.5, ~100 tokens/call)	eval/invoice_summary.json · agent/tone_guard.py	\$0.002/call
Expected regeneration (30% retry rate × \$0.002)	eval/invoice_summary.json · method/ablation_results.json	\$0.0006/lead
Total per qualified lead	eval/invoice_summary.json	\$0.015 — hard cap: \$5.00

Rig cost: \$0.00. All simulations on provisioned challenge-week keys. Kill-switch unset → \$0 Resend + \$0 AT charges. Source: eval/invoice_summary.json

3. SPEED-TO-LEAD — STALLED-THREAD RATE

Stalled defined: No outbound agent action within 4 hours of an inbound prospect event — measurable from data/threads.json timestamps.

Condition	Stall rate	Response latency	Source
Tenacious manual (CEO/CFO estimate)	30–40% stalled >4 h	2–4 h partner recall delay	Internal estimate
System — challenge-week (n=3 events)	0/3 = 0% stalled	p50: 4.0 s · max: 4.1 s	data/threads.json · eval/trace_log.jsonl

Caveat: n=3 (simulated webhooks only — kill-switch active). 0% is structurally guaranteed by synchronous dispatch in agent/email_handler.py, not a statistical result. Projected live stall rate $\leq 20\%$ (40% manual × 50% context-driven = 20%). Requires 30-day pilot to confirm.

4. COMPETITIVE-GAP OUTBOUND PERFORMANCE

Variant	Sends (week)	Signal injected	Replies	Benchmark projection
Full method (non-abstained)	10 (staff sink)	100% — hiring_brief + gap_brief	3 simulated; organic: 0	7–12% (Woodpecker 2024)
Baseline (generic pitch)	0 — not sent	0% — no signal injection	not measurable	2–4% (Woodpecker 2024)
Abstained (confidence <0.40)	~18% of triggers	Generic only	—	—

Caveat — no live A/B comparison: The baseline had zero sends; no reply rate was measured for it. The delta (7–12% vs. 2–4%) is not a direct system comparison — it is two Woodpecker 2024 benchmark bands applied independently. The structural difference is confirmed by code inspection: agent/llm_composer.py always injects hiring_brief + gap_brief (full method) vs. template with zero injection (baseline). n≈200 organic sends per variant required for empirical measurement. Source: agent/icp_classifier.py · data/threads.json

5. PILOT SCOPE RECOMMENDATION

Segment: Segment 1 — Recently Funded Series A/B (classifier confidence 0.84, lowest offshore-perception risk). **Volume:** 50 leads/week × 4 weeks = 200 leads; SDR reviews each draft (~2 h/week). **Budget:** \$3/month LLM cost; kill-switch unset until SDR approval gate is wired. **Success criterion:** ≥8 discovery calls booked in 30 days (4% meeting rate). If met: expand to Segment 3 at 100 leads/week.

6. FOUR FAILURE MODES T²-BENCH DOES NOT CAPTURE

Failure	What it is	Why benchmark misses it	What would catch it	Cost to add
Bench over-commitment	Agent commits "3 Go engineers in 2 weeks" when bench_summary.json is 48–72 h stale and the stack is fully engaged	τ^2 -Bench retail has no bench capacity state; no capacity-constrained tasks	P-009–P-012 (built): inject stale bench file, ask for capacity, check response vs. actual counts. Needs weekly bench freshness automation.	1 eng-day; \$0.20/run
Offshore-perception language	Segment 2 pitch uses "replace higher-cost roles" framing to a CTO with a prior failed offshore engagement — triggers complaint or delete	τ^2 -Bench users have no offshore-sensitivity backstory; no cultural context injected into persona	Persona-augmented probes: inject CTO with offshore-failure flag; score whether agent avoids cost-arbitrage language. P-013–P-015 partial.	2 eng-days; \$0.50/run
ICP conflict signals (P-004)	Layoff 90d ago + new CTO 45d ago; agent references both signals in one email — reads as researched-but-confused, aggressive	τ^2 -Bench tasks have single clean user intents; no overlapping conflicting business signals in prospect state	Multi-signal conflict probes: seed two overlapping signals, verify agent picks exactly one and suppresses the secondary. Requires honesty_flags check in prompt.	1 eng-day; \$0.15/run
Multi-thread leakage	Two SDRs contact CEO and CTO of same company in the same week with different segment pitches — contradictory framing (\$108K deal risk)	τ^2 -Bench is single-agent single-session; no shared state across parallel agent instances	Parallel-thread probes: two agent instances vs. same company_id in thread_store; verify second agent detects active thread and suppresses. P-016–P-018 cover this.	3 eng-days; \$0.40/run

7. PUBLIC-SIGNAL LOSSINESS — AI MATURITY SCORING

Profile	What the system sees	What the agent does wrong	Business impact
Quietly sophisticated / publicly silent (fintech, proprietary ML stack, careers behind login)	0 AI roles detected, no named ML leadership, no OSS → ai_maturity_score=0 , weak_signal=true	Abstains or sends generic opener. Prospect (actually AI-mature) reads it as uninformed outreach and ignores it.	False negative: missed warm lead. \$5,400 expected loss/event (15% close × \$36K ACV). Est. 12–18% of AI-mature firms in closed-career sectors.
Loud but shallow (Series A, "AI-first" press releases, 0 shipped ML product)	High exec commentary + funding event + 2–3 AI buzzword titles → ai_maturity_score=2 , confidence=0.78	Sends signal-grounded email citing "MLOps gap vs. Tecton" — but prospect has no ML platform. Pitch lands confused; CTO forwards as vendor-noise example.	False positive: brand damage. \$500 rep cost × 5% rate × 1,000 emails = \$25,000. Triggers kill-switch Threshold 3 (>3 wrong-signal HubSpot flags/7d).

8. ONE HONEST UNRESOLVED FAILURE — P-004

P-004: Layoff + New CTO (Conflicting Signals) — Residual trigger rate 3/10 after fix

What happens: Layoff 90d ago (Segment 2) + new CTO 45d ago (Segment 3). Classifier picks Segment 3 on recency, but LLM email references both events — reads as "we know you laid people off and got a new boss." Aggressive and off-putting.

Why unresolved: honesty_flags now flags conflicting_segment_signals, but the system prompt lacks a suppression rule. Fix: "If honesty_flags includes conflicting_segment_signals, reference only the primary_segment_match signal and suppress the secondary."

Impact if deployed: ~8% of Seg 2/3 prospects have overlapping signals → ~4 conflicted leads/week at 50/week. Cost: \$500 rep + \$18K expected deal loss each. **~\$72,500/week unresolved. Do not expand past Segment 1 without this fix.**